23. The electric field in a region is given by

$$\vec{E} = (10x + 4)\,\hat{i}$$

where x is in m and E is in N/C. Calculate the amount of work done in taking a unit charge from

(i) (5 m, 0) to (10 m, 0)

(ii) (5 m, 0) to (5 m, 10 m) **3**

31. (a) (i) Obtain the expression for the capacitance of a parallel plate capacitor with a dielectric medium between its plates.

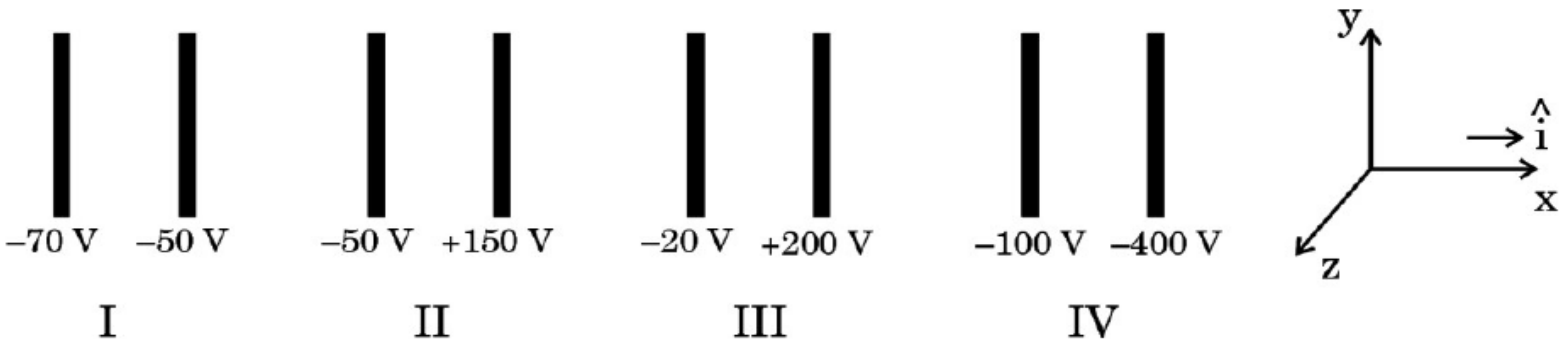
(ii) A charge of 6 μC is given to a hollow metallic sphere of radius 0·2 m. Find the potential at (i) the surface and (ii) the centre of the sphere. *5*

OR

(b) (i) A charge + Q is placed on a thin conducting spherical shell of radius R. Use Gauss's theorem to derive an expression for the electric field at a point lying (i) inside and (ii) outside the shell.

(ii) Show that the electric field for same charge density (σ) is twice in case of a conducting plate or surface than in a nonconducting sheet. *5*

29. The figure shows four pairs of parallel identical conducting plates, separated by the same distance 2·0 cm and arranged perpendicular to x-axis. The electric potential of each plate is mentioned. The electric field between a pair of plates is uniform and normal to the plates.



(i) For which pair of the plates is the electric field $\vec{E}$ along $\hat{i}$? *1*

(A) I (B) II

(C) III (D) IV

(ii) An electron is released midway between the plates of pair IV. It will : *1*

(A) move along $\hat{i}$ at constant speed

(B) move along $-\hat{i}$ at constant speed

(C) accelerate along $\hat{i}$

(D) accelerate along $-\hat{i}$

(iii) Let V_0 be the potential at the left plate of any set, taken to be at x = 0 m. Then potential V at any point ($0 \leq x \leq 2$ cm) between the plates of that set can be expressed as : *1*

(A) $V = V_0 + \alpha x$ (B) $V = V_0 + \alpha x^2$

(C) $V = V_0 + \alpha x^{1/2}$ (D) $V = V_0 + \alpha x^{3/2}$

where α is a constant, positive or negative.

(iv) (a) Let E_1, E_2, E_3 and E_4 be the magnitudes of the electric field between the pairs of plates, I, II, III and IV respectively. Then : *1*

(A) $E_1 > E_2 > E_3 > E_4$ (B) $E_3 > E_4 > E_1 > E_2$

(C) $E_4 > E_3 > E_2 > E_1$ (D) $E_2 > E_3 > E_4 > E_1$

OR

(b) An electron is projected from the right plate of set I directly towards its left plate. It just comes to rest at the plate. The speed with which it was projected is about :
(Take (e/m) $= 1{\cdot}76 \times 10^{11}$ C/kg) *1*

(A) $1{\cdot}3 \times 10^5$ m/s (B) $2{\cdot}6 \times 10^6$ m/s

(C) $6{\cdot}5 \times 10^5$ m/s (D) $5{\cdot}2 \times 10^7$ m/s

1. The plates P_1 and P_2 of a 2 μF capacitor are at potentials 25 V and – 25 V respectively. The charge on plate P_1 will be :

(A) 0·02 mC

(B) 0·1 mC

(C) 0·1 μC

(D) 1 μC

1. An isolated conductor, with a cavity, has a net charge $+Q$. A point charge $+q$ is inside the cavity. The charges on the cavity wall and the outer surface are respectively :

(A) 0 and Q

(B) $-q$ and $Q - q$

(C) $-q$ and $Q + q$

(D) 0 and $Q - q$

2. A proton is taken from point P_1 to point P_2, both located in an electric field. The potentials at points P_1 and P_2 are -5 V and $+5$ V respectively. Assuming that kinetic energies of the proton at points P_1 and P_2 are zero, the work done on the proton is :

(A) $-1{\cdot}6 \times 10^{-18}$ J

(B) $1{\cdot}6 \times 10^{-18}$ J

(C) Zero

(D) $0{\cdot}8 \times 10^{-18}$ J

1. Consider a group of charges q_1, q_2, q_3 … such that $\Sigma q \neq 0$. Then equipotentials at a large distance, due to this group are approximately :

(A) Plane (B) Spherical surface

(C) Paraboloidal surface (D) Ellipsoidal surface

32. (a) (i) Obtain an expression for the electric potential due to a small dipole of dipole moment $\vec{p}$, at a point $\vec{r}$ from its centre, for much larger distances compared to the size of the dipole.

(ii) Three point charges q, 2q and nq are placed at the vertices of an equilateral triangle. If the potential energy of the system is zero, find the value of n. *5*

1. The capacitance of a parallel plate capacitor having a medium of dielectric constant K = 4 in between the plates is C. If this medium is removed, then the capacitance of the capacitor becomes :

(A) 4C

(B) C

(C) $\dfrac{C}{4}$

(D) 2C

32. (a) (i) Derive an expression for potential energy of an electric dipole $\vec{p}$ in an external uniform electric field $\vec{E}$. When is the potential energy of the dipole (1) maximum, and (2) minimum ?

(ii) An electric dipole consists of point charges – 1·0 pC and + 1·0 pC located at (0, 0) and (3 mm, 4 mm) respectively in x – y plane. An electric field $\vec{E} = \left(\frac{1000\ V}{m}\right)\hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole. *5*

OR

(b) (i) An electric dipole (dipole moment $\vec{p} = p\,\hat{i}$), consisting of charges $-q$ and q, separated by distance $2a$, is placed along the x-axis, with its centre at the origin. Show that the potential V, due to this dipole, at a point x, $(x >> a)$ is equal to $\frac{1}{4\pi\varepsilon_0} \cdot \frac{\vec{p} \cdot \hat{i}}{x^2}$.

(ii) Two isolated metallic spheres S_1 and S_2 of radii 1 cm and 3 cm respectively are charged such that both have the same charge density $\left(\frac{2}{\pi} \times 10^{-9}\right) C/m^2$. They are placed far away from each other and connected by a thin wire. Calculate the new charge on sphere S_1. *5*

22. Three point charges Q_1, Q_2 and Q_3 are located in x – y plane at points (– d, 0), (0, 0) and (d, 0) respectively. Q_1 and Q_3 are identical and Q_2 is positive. What will be the nature and value of Q_1 so that the potential energy of the system is zero ? *3*

2. Ten capacitors, each of capacitance 1 μF, are connected in parallel to a source of 100 V. The total energy stored in the system is equal to :

(A) 10^{-2} J

(B) 10^{-3} J

(C) $0{\cdot}5 \times 10^{-3}$ J

(D) $5{\cdot}0 \times 10^{-2}$ J

31. (a) (i) A capacitor of capacitance C is charged to a potential difference V by a battery. The battery is disconnected and the separation between the plates is halved. How will the following quantities be affected ?

(1) Capacitance of the capacitor

(2) Electric field between the plates

(3) Energy stored in the capacitor

Justify your answer in each case.

(ii) A capacitor of 20 μF is charged to 30 V. It is then connected to an uncharged 30 μF capacitor. Calculate the charges on each capacitor in equilibrium. *5*

OR

(b) (i) Obtain an expression for electric potential at a distance r from a point charge Q.

(ii) Two point charges of 10 μC and – 5 μC are placed at (– 3 cm, 0) and (6 cm, 0) in an external electric field $E = \frac{A}{r^2}$, where $A = 1{\cdot}8 \times 10^5\ N\ C^{-1}\ m^2$. Calculate the electrostatic energy of the system. *5*

(iv) (b) A electric dipole consists of two point charges $+ 2$ pC and $- 2$ pC, separated by a distance 1 mm. It is kept in a uniform electric field of 2×10^5 NC^{-1}. The amount of work done in rotating it from its position of stable to unstable equilibrium will be : *1*

(A) 2×10^{-10} J

(B) 4×10^{-10} J

(C) 8×10^{-10} J

(D) $1{\cdot}6 \times 10^{-11}$ J

(iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be : *1*

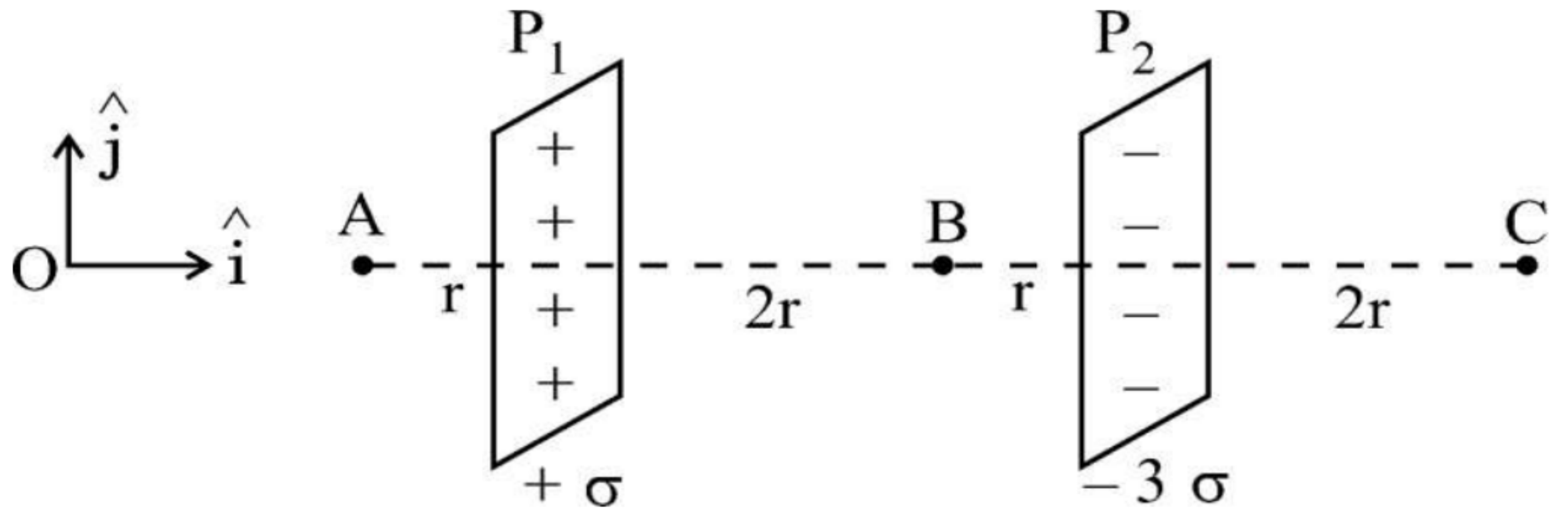
(A) p (B) 2p

(C) $p\sqrt{2}$ (D) $\frac{p}{\sqrt{2}}$

8. Two particles A and B of the same mass but having charges q and 4q respectively, are accelerated from rest through different potential differences V_A and V_B such that they attain same kinetic energies. The value of $\left(\frac{V_A}{V_B}\right)$ is :

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 2 (D) 4

(b) Two large plane sheets P_1 and P_2 having charge densities $+\sigma$ and -3σ respectively are arranged parallel to each other as shown in the figure. Find the net electric field ($\vec{E}$) at points A, B and C. *3*



23. Two point charges of 10 μC and 20 μC are located at points (– 4 cm, 0, 0) and (5 cm, 0, 0) respectively, in a region with electric field $E = \frac{A}{r^2}$, where $A = 2 \times 10^6 \ NC^{-1} \ m^2$ and $\vec{r}$ is the position vector of the point under consideration. Calculate the electrostatic potential energy of the system. *3*

1. Three points charges + q, – 4q and + 2q are kept at the vertices of an equilateral triangle of side ‘a’. The electrostatic potential energy of this system of charges is :

(A) $-\dfrac{1}{4\pi\,\varepsilon_0}\left(\dfrac{q^2}{a^2}\right)$

(B) $+\dfrac{1}{4\pi\,\varepsilon_0}\left(\dfrac{8q^2}{a^2}\right)$

(C) $+\dfrac{1}{4\pi\,\varepsilon_0}\left(\dfrac{6q^2}{a}\right)$

(D) $-\dfrac{1}{4\pi\,\varepsilon_0}\left(\dfrac{10q^2}{a}\right)$

25. Explain the polarization of a dielectric medium filled inside a charged capacitor. How does the capacity of a capacitor increase due to filling ? What is dielectric constant of a dielectric ? *3*

29. The electric potential at a point in an electric field is the work done in bringing a unit positive charge from infinity to this point. If the potential difference between any two points at a surface is zero, the surface is called an equipotential surface. The shape of an equipotential surface can give direction of electric field at a point on it. It may be a closed surface or an open surface, depending on the charges creating the electric field.

(i) The shape of equipotential surface due to an isolated point charge is : *1*

(A) ellipsoidal, with charge at its one foci
(B) plane surface not passing through point charge
(C) spherical with charge at its centre
(D) cylindrical with charge at its axis

(ii) The equipotential surfaces in a uniform electric field acting along + Z direction are : *1*

(A) planes parallel to the XY plane
(B) concentric spherical surfaces
(C) planes parallel to the YZ plane
(D) planes parallel to the XZ plane

(iii) Which of the following statements is ***not*** true for equipotential surfaces ? *1*

(A) Potentials at two equipotential surfaces are different.

(B) No work is done in moving a charge on an equipotential surface.

(C) Equipotential surfaces always enclose some volume.

(D) Two equipotential surfaces do not intersect each other.

(iv) (a) The angle between the electric field at a point and outward normal at that point on an equipotential surface is : *1*

(A) 90° (B) 45°

(C) 30° (D) 0°

OR

(iv) (b) A and B are two equipotential surfaces around a point charge Q. Surface A is closer to the charge than surface B. If a test charge is released between them, it will : *1*

(A) remain stationary

(B) move from A to B

(C) move from B to A

(D) move around Q in a circular path between the equipotential surfaces A and B

(b) (i) Obtain an expression for the capacitance of a parallel plate capacitor.

(ii) Two identical plates, each of area A having surface charge densities $+\sigma$ and $-\sigma$, respectively, are separated by a distance d in air and a dielectric of dielectric constant K is filled between them. Write the expressions for : *5*

(1) the potential difference between the plates

(2) capacitance of the capacitor so formed

(3) energy stored in this capacitor